

3.3 Statements with Multiple Quantifiers (Homework)

 INSTRUCTOR

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 REQUEST EXTENSION

Current Score: – / 100 POINTS | 0.0 %

Scoring and Assignment Information

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QUESTION	1	2	3	4	5	6	7	8	9
POINTS	– / 11.11	– / 11.11	– / 11.11	– / 11.11	– / 11.11	– / 11.11	– / 11.11	– / 11.11	– / 11.12

Assignment Submission

For this assignment, you submit answers by question parts. The number of submissions remaining for each question part only changes if you submit or change the answer.

Assignment Scoring

Your last submission is used for your score.

1. **[– / 11.11 Points]**

DETAILS

MY NOTES

PRACTICE ANOTHER

EppDiscMath5 3.3.004.

The following statement is true.

$\forall$ real number x , $\exists$ an integer n such that $n > x$.

For each value of x given below, fill in a value of n to make the predicate " $n > x$ " true.

(a) $x = 13.89$

(a) $x = 10^{10}$
Let $n =$

(b) $x = 10^9$
Let $n =$

(c) $x = 5^{5^5}$
Let $n =$

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2. [- / 11.11 Points]

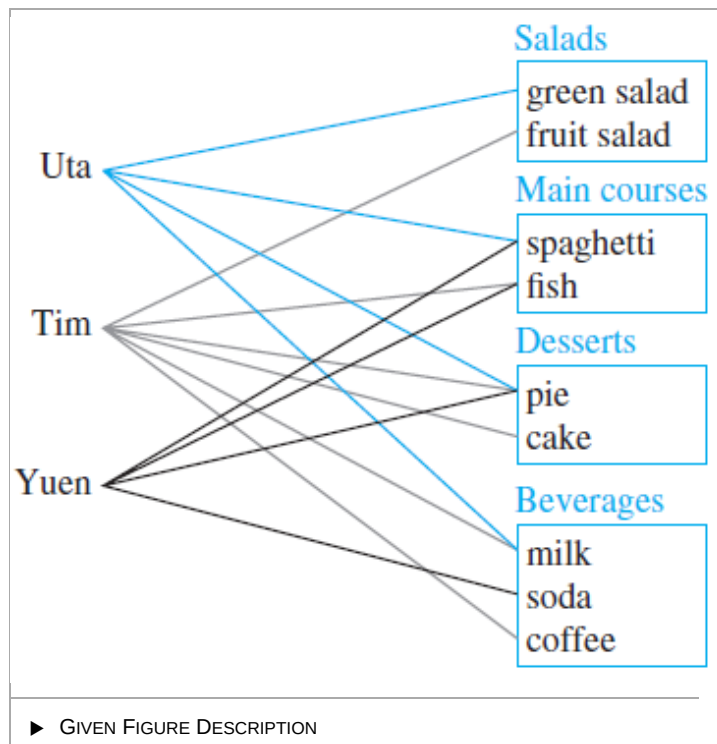
DETAILS

MY NOTES

EppDiscMath5 3.3.010.

Use the figure below to determine whether the statements in (a)–(f) are true or false.

GIVEN FIGURE



(a) $\forall$ student S , $\exists$ a dessert D such that S chose D .

- ☐ True. Every student chose at least one dessert: Uta chose pie, Tim chose both pie and cake, and Yuen chose pie.
- ☐ True. Although some students didn't choose dessert, there existed an option for them to choose from.
- ☐ False. Uta did not have dessert.
- ☐ False. Tim had more than one dessert.
- ☐ False. None of the students chose a beverage.

☐ False. None of the students chose dessert.

(b) $\forall$ student S , $\exists$ a salad T such that S chose T .

- ☐ True. Every student chose at least one salad: Uta chose green salad, Tim chose fruit salad, and Yuen chose green salad.
- ☐ True. Although some students didn't choose salad, there existed an option for them to choose from.
- ☐ False. Yuen did not have a salad.
- ☐ False. Tim had more than one salad.
- ☐ False. None of the students had a salad.

(c) $\exists$ a dessert D such that $\forall$ student S , S chose D .

- ☐ True. Every student chose pie.
- ☐ True. Although some students didn't choose dessert, there existed an option for them to choose from.
- ☐ False. Uta did not have dessert.
- ☐ False. Tim had more than one dessert.
- ☐ False. None of the students chose dessert.

(d) $\exists$ a beverage B such that $\forall$ student D , D chose B .

- ☐ True. Every student chose soda.
- ☐ True. Although some students didn't choose a beverage, there existed an option for them to choose from.
- ☐ False. Uta did not have a beverage.
- ☐ False. Tim only had one beverage.
- ☐ False. There was no particular beverage chosen by every student.

(e) $\exists$ an item I such that $\forall$ student S , S did not choose I .

- ☐ True. Every student chose every available item on the menu.
- ☐ True. Although some students didn't choose a certain item from the menu, there existed an option for them to choose from.
- ☐ False. Uta did not have a beverage.
- ☐ False. Tim only had one dessert.
- ☐ False. Every item was chosen by at least one student.

(f) $\exists$ a station Z such that $\forall S, \exists$ an item I such that S chose I from Z .

- ☐ True. Every student chose a main course, every student chose a dessert, and every student chose a beverage.
- ☐ True. Although some students didn't choose a food from some of the stations, there existed an option for them to choose from.
- ☐ False. Uta did not have a beverage.
- ☐ False. Tim only had one dessert.
- ☐ False. None of the items were chosen from any station by any of the students.

Resources

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3. [- / 11.11 Points]

DETAILS

MY NOTES

EppDiscMath5 3.3.012.

Let $D = E = \{-2, -1, 0, 1, 2\}$. Write negations for each of the following statements and determine which is true, the given statement or its negation.

(a) $\forall x \text{ in } D, \exists y \text{ in } E \text{ such that } x + y = 1.$

READ AND COMPLETE THE FOLLOWING.

Negation: in D such that in $E, x + y$ 1.

Which is true?

- ☐ The given statement is true.
- ☐ The negation is true.

(b) $\exists x \text{ in } D \text{ such that } \forall y \text{ in } E, x + y = -y.$

READ AND COMPLETE THE FOLLOWING.

Negation: in D , in E such that $x + y$ $-y$.

Which is true?

- ☐ The given statement is true.
- ☐

☐ The negation is true.

(c) $\forall x \text{ in } D, \exists y \text{ in } E \text{ such that } xy \geq y.$

READ AND COMPLETE THE FOLLOWING.

Negation: in D such that in E, xy $y.$

Which is true?

☐ The given statement is true.

☐ The negation is true.

(d) $\exists x \text{ in } D \text{ such that } \forall y \text{ in } E, x \leq y.$

READ AND COMPLETE THE FOLLOWING.

Negation: in D , in E such that x $y.$

Which is true?

☐ The given statement is true.

☐ The negation is true.

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4. [- / 11.11 Points]

DETAILS

MY NOTES

EppDiscMath5 3.3.017.

Consider the following statement.

$$\forall r \in \mathbf{Q}, \exists \text{ integers } a \text{ and } b \text{ such that } r = \frac{a}{b}.$$

Choose the answer that correctly expresses the statement in English without the use of symbols or variables.

☐ Every real number is equal to a ratio of some two whole numbers.

☐ Every rational number is equal to a ratio of some two integers.

☐ There is at least one real number that is not equal to a ratio of any two whole numbers.

☐ Every rational number is equal to a ratio of some two real numbers.

- ☐ Every rational number is equal to a ratio of some two real numbers.
- ☐ There is at least one rational number that is not equal to a ratio of any two integers.

Which of the following is a negation for the statement?

- ☐ Every rational number is equal to a ratio of some two integers.
- ☐ There is at least one real number that is not equal to a ratio of any two whole numbers.
- ☐ There is at least one rational number that is not equal to a ratio of any two real numbers.
- ☐ Every real number is equal to a ratio of some two whole numbers.
- ☐ There is at least one rational number that is not equal to a ratio of any two integers.

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5. [- / 11.11 Points]

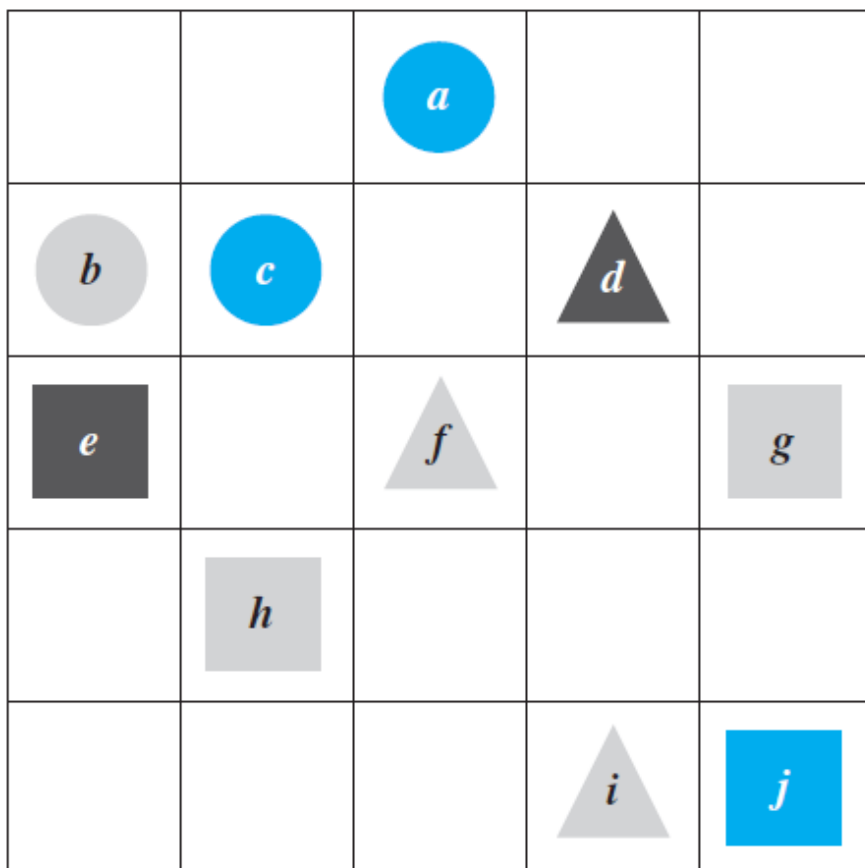
DETAILS

MY NOTES

EppDiscMath5 3.3.020.

Recall that reversing the order of the quantifiers in a statement with two different quantifiers may change the truth value of the statement—but it does not necessarily do so. Both pairs of statements in (a) and (b) below refer to the Tarski world introduced in Example 3.3.1 and shown again below. In each pair, the order of the quantifiers is reversed, but everything else is the same.

GIVEN FIGURE



For each pair, determine whether the statements have the same or opposite truth values.

- (a) (1) For every square y there is a triangle x such that x and y have different colors.
 (2) There is a triangle x such that for every square y , x and y have different colors.

For each pair, determine whether the statements have the same or opposite truth values.

- ☐ Only statement 1 is true.
- ☐ Only statement 2 is true.
- ☐ Both statements are true.
- ☐ Neither of the statements is true.

- (b) (1) For every circle y there is a square x such that x and y have the same color.
 (2) There is a square x such that for every circle y , x and y have the same color.

For each pair, determine whether the statements have the same or opposite truth values.

- ☐ Only statement 1 is true.
- ☐ Only statement 2 is true.
- ☐ Both statements are true.
- ☐ Neither of the statements is true.

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6. [- / 11.11 Points]

DETAILS

MY NOTES

EppDiscMath5 3.3.023.

For each statement in (a) and (b), choose the answer that expresses the statement without the use of variables or the symbol $\forall$ or $\exists$. Then indicate whether the statement is true or false.

- (a) $\forall$ nonzero real number r , $\exists$ a real number s such that $rs = 1$.

- ☐ Given any real number, a real number can be found so that the sum of the two equals 1.
- ☐ Given any nonzero real number, a real number cannot be found so that the product of the two equals 0.
- ☐ There is a real number whose sum with every real number equals 1.
- ☐ Given any nonzero real number, a real number can be found so that the product of the two equals 1.
- ☐ There is a real number whose product with every nonzero real number equals 1.

Is the statement true or false?

Is the statement true or false?

☐ True

☐ False

(b) $\exists$ a real number r such that $\forall$ nonzero real numbers s , $rs = 1$.

- ☐ Given any real number, a real number can be found so that the sum of the two equals 1.
- ☐ There is no real number whose product with every nonzero real number equals 0.
- ☐ Given any nonzero real number, a real number can be found so that the product of the two equals 1.
- ☐ There is a real number whose product with every nonzero real number equals 1.
- ☐ There is a real number whose sum with every real number equals 1.

Is the statement true or false?

☐ True

☐ False

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7. [- / 11.11 Points]

DETAILS

MY NOTES

EppDiscMath5 3.3.030.

Consider the following statement.

$\exists x \in \mathbf{R}$ such that $\forall y \in \mathbf{R}^-$ (the set of negative real numbers), $x > y$.

(a) If the symbols $\forall$ and $\exists$ are interchanged, what is the result?

Statement with interchanged quantifier symbols:

READ AND COMPLETE THE FOLLOWING.

$\in \mathbf{R}$, $\in \mathbf{R}^-$ such that $x > y$.

(b) Select which is true.

- ☐ The given statement is true.
- ☐ The new statement with interchanged quantifiers is true.

- ☐ Neither statement is true.
- ☐ Both statements are true.

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8. [- / 11.11 Points]

DETAILS

MY NOTES

PRACTICE ANOTHER

EppDiscMath5 3.3.036.

Consider the following statement.

Somebody trusts everybody.

(a) Which of the following expresses the statement formally using quantifiers and variables?

- ☐ $\exists$ a person x such that $\forall$ person y , x does not trust y .
- ☐ $\forall$ person x , $\exists$ a person y such that x does not trust y .
- ☐ $\exists$ a person x such that $\forall$ person y , x trusts y .
- ☐ $\forall$ person x , $\exists$ a person y such that x trusts y .

(b) Which of the following is a negation for the statement?

- ☐ $\exists$ a person x such that $\forall$ person y , x does not trust y .
- ☐ $\forall$ person x , $\exists$ a person y such that x does not trust y .
- ☐ $\exists$ a person x such that $\forall$ person y , x trusts y .
- ☐ $\forall$ person x , $\exists$ a person y such that x trusts y .

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9. [- / 11.12 Points]

DETAILS

MY NOTES

PRACTICE ANOTHER

EppDiscMath5 3.3.038.

Consider the following statement.

Every action has an equal and opposite reaction.

Every action has an equal and opposite reaction.

(a) Which of the following expresses the statement formally using quantifiers and variables?

- ☐ $\forall$ action A , $\exists$ an opposite action to R such that R is equal to A .
- ☐ $\forall$ action A , $\exists$ a reaction R such that R is equal and opposite to A .
- ☐ $\exists$ an action A such that $\forall$ reaction R , R is not equal to A or is not opposite to A .
- ☐ $\exists$ an opposite action A such that $\forall$ reaction R , R is not equal to A .

(b) Which of the following is a negation for the statement?

- ☐ $\forall$ action A , $\exists$ an opposite action to R such that R is equal to A .
- ☐ $\forall$ action A , $\exists$ a reaction R such that R is equal and opposite to A .
- ☐ $\exists$ an action A such that $\forall$ reaction R , R is not equal to A or is not opposite to A .
- ☐ $\exists$ an opposite action A such that $\forall$ reaction R , R is not equal to A .

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